On the initial value problem in Lovelock and Horndeski theories of gravity

GP and Harvey Reall, arXiv:1705.04370
GP, arXiv:1710.10155

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HECAP Seminar, ICTP, Trieste
March 1st, 2018

General Relativity

General Relativity provides a very elegant and simple mathematical description of the gravitational interaction.

Many successful experimental verifications (bending of light, Shapiro time delay, precession of Mercury, indirect and direct observation of gravitational waves . . .)

Modifying GR?

Despite the successes of GR, some phenomena do not have a satisfactory explanation.

- Singularities
- Quantum gravity?
- Dark Universe

It may be necessary to modify the theory!

Lovelock theorem

Lovelock's theorem: in $d = 4$, the Einstein tensor is the unique rank-2 tensor which is

- symmetric;
- divergence free;
- depends only on g, ∂g and $\partial^2 g$.

In other words, the most general diffeomorphism-covariant metric theory of gravity in $d = 4$ with second order equations of motion is GR.

$$G_{ab} + \Lambda g_{ab} = 8\pi T_{ab}$$

How to modify GR?

If we require the theory to arise from a diffeo-invariant action and to have second order equations of motion (cf. Ostrogradsky instabilities), Lovelock theorem tells us that we can

- consider higher dimensional spacetimes,
- or consider additional degrees of freedom (e.g. additional scalar field).

Lovelock theories

If $d > 4$ but other assumptions are satisfied $\rightarrow$ most general theory is *Lovelock gravity*

$$A_{ab} = 8\pi T_{ab}$$

where

$$A^a{}_b = G^a{}_b + \Lambda \delta^a_b + \sum_{p \geq 2} k_p \delta^{ac_1 \ldots c_{2p}}_{bd_1 \ldots d_{2p}} R_{c_1 c_2}{}^{d_1 d_2} \ldots R_{c_{2p-1} c_{2p}}{}^{d_{2p-1} d_{2p}}$$

Remark: antisymmetry ensures that the sum is finite.
In $d = 4$ they reduce to GR.

Horndeski theories

Horndeski theories are the most general 4-dim diffeo-covariant theories of gravity coupled to a scalar field Φ with second order EoMs.

$$S = \frac{1}{16\pi G}\int d^4x\sqrt{-g}(\mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4 + \mathcal{L}_5)$$

where

$$\mathcal{L}_1 = R + X - V(\Phi) \qquad \mathcal{L}_2 = \mathcal{G}_2(\Phi, X) \qquad \mathcal{L}_3 = \mathcal{G}_3(\Phi, X)\Box\Phi$$

$$\mathcal{L}_4 = \mathcal{G}_4(\Phi, X)R + \partial_X\mathcal{G}_4(\Phi, X)\delta^{ac}_{bd}\,\nabla_a\nabla^b\Phi\,\nabla_c\nabla^d\Phi$$

$$\mathcal{L}_5 = \mathcal{G}_5(\Phi, X)G_{ab}\nabla^a\nabla^b\Phi - \frac{1}{6}\partial_X\mathcal{G}_5(\Phi, X)\delta^{ace}_{bdf}\,\nabla_a\nabla^b\Phi\,\nabla_c\nabla^d\Phi\,\nabla_e\nabla^f\Phi$$

and we have defined $X = -\frac{1}{2}(\nabla\Phi)^2$.

Horndeski theories

Many theories of interest in cosmology/astrophysics belong to this class, e.g.:

- *k-essence*;
- *Galileons*;
- *Brans-Dicke*;
- *Einstein–dilaton–Gauss-Bonnet*.

Aim

Question: are Lovelock or Horndeski theories viable alternatives to Einstein's GR?

We will study their mathematical properties in order to answer this questions.

Cauchy problem

Essential requirement for a theory to make sense: initial value problem must be (locally) *well-posed*.

Well-posedness: given initial data, a solution *exists*, is *unique* and it *continuously depends on the data*.

GR is well-posed (*Choquet-Bruhat 1952*).

Not known whether Lovelock or Horndeski admit a well-posed Cauchy problem.
Remark: if data is analytic $\rightarrow$ Cauchy-Kovalevskaya thm (*Choquet-Bruhat 1988*)

Cauchy problem

Well-posedness of full non-linear problem $\rightarrow$ hard!

Consider problem of non-linear perturbations of a background $u^{(0)}$. For this to be well-posed, necessary that all initial value problems obtained by linearising around any background in a neighbourhood of $u^{(0)}$ be well-posed.

High freq problem $\rightarrow$ look at small neighbourhood $\rightarrow$ constant coefficients problem.

Well-posedness of such linearised problems related to concepts of *weak* and *strong hyperbolicity*.

Cauchy problem

Consider linearisation of a system of 2nd order PDEs (with constant coeffs) around background $u^{(0)}$

$$P^{\mu\nu IJ}(u^{(0)})\partial_\mu\partial_\nu u_J + \ldots = 0$$

Spatial Fourier transform

$$A^{IJ}\partial_t^2\tilde{u}_J + B^{IJ}(\xi_i)\partial_t\tilde{u}_J + C^{IJ}(\xi_i)\tilde{u}_J + \ldots = 0$$

where

$$A^{IJ} = P^{00IJ} \qquad B^{IJ}(\xi_i) = 2\xi_i P^{(0i)IJ} \qquad C^{IJ}(\xi_i) = \xi_i\xi_j P^{ijIJ}$$

Rewrite as first order (in time) system

$$\partial_t\tilde{w} = i\mathcal{M}(\xi_i)\tilde{w} \qquad \tilde{w}^T = \left(\sqrt{1+|\xi|^2}\tilde{u}, -i\partial_t\tilde{u}\right)$$

Cauchy problem

Formal solution is given by

$$\tilde{w}(t, \xi_i) = \exp(i\mathcal{M}(\xi_i)t)\tilde{w}(0, \xi_i)$$

and hence

$$w(t, x) = \frac{1}{(2\pi)^{d-1}} \int d^{d-1}\xi \exp(-i\xi_i x^i) \exp(i\mathcal{M}(\xi_i)t)\tilde{w}(0, \xi_i)$$

Need to ensure convergence as $|\xi| \to \infty$. Guaranteed if

$$|| \exp(i\mathcal{M}(\xi_i)t)|| \leq f(t)$$

for some continuous f indept of ξ_i.

Cauchy problem

If this holds then

$$||w||(t) \leq f(t)||w||(0)$$

which implies local well-posedness of the Cauchy problem.

Let w_1 and w_2 be two solutions arising from data w_1^0, w_2^0 respectively. The above implies

$$||w_1 - w_2||(t) \leq f(t)||w_1^0 - w_2^0||$$

i.e. continuous dependence on initial data.
Letting $w_1^0 = w_2^0$ proves uniqueness.

Cauchy problem

The convergence of the integral is a *high-frequency* question. Let $t' = |\xi| t$, $\xi_i = \hat{\xi}_i |\xi|$ and take the limit $|\xi| \to \infty$ at fixed t'

$$|| \exp(i\mathcal{M}(\xi_i)t)|| \leq f(t) \quad \to \quad || \exp(iM(\hat{\xi}_i)t')|| \leq k \equiv f(0)$$

where M is the *principal part* of $\mathcal{M}$

$$M(\xi_i) = \begin{pmatrix} 0 & I \\ -A^{-1}C(\xi_i) & -A^{-1}B(\xi_i) \end{pmatrix}$$

What conditions must M satisfy for well-posedness to hold?

Weak hyperbolicity

A necessary condition is that its eigenvalues be *real* $\rightarrow$ *weak hyperbolicity.*

Why? Otherwise the integrand grows exponentially in $|\xi|$.

In other words: weak hyperbolicity means that no solution grows exponentially in time, arbitrarily fast at arbitrarily short spatial wavelengths

Strong hyperbolicity

This condition is NOT *sufficient* to ensure well-posedness.

Suppose M has real eigenvalues but not diagonalisable.
In Jordan form there is at least a non trivial block.

Exponentiating $\rightarrow$ terms growing as $|\xi|^p$ in FT integral.

Remark: can still have linear well-posedness, but depends on form of lower order terms.
At best: energy estimate with loss of derivatives, e.g.

$$||w||(t) \leq f(t)(||w||(0) + ||\partial w||(0))$$

Likely fatal for non-linear well-posedness.

Weak/Strong hyperbolicity - summary

Weak hyperbolicity: $M(\xi_i)$ has only real eigenvalues;

Strong hyperbolicity: $M(\xi_i)$ is diagonalisable (with a smooth, bounded symmetrizer).

Weak/Strong hyperbolicity - summary

Define the *principal symbol*:

$$P(\xi)^{IJ} = P^{\mu\nu IJ}\xi_\mu\xi_\nu$$

A 1-form ξ is characteristic iff $\det P(\xi) = 0$
(e.g. in GR, ξ characteristic iff null).

Weak hyperbolicity: any characteristic 1-form with real spatial components has real time component.

Gauge fixing

GR, Lovelock and Horndeski are diffeo-covariant theories $\rightarrow$ need to fix gauge to establish hyperbolicity!

Example: the Einstein equations are *strongly* hyperbolic in (generalised) harmonic gauge, but only *weakly* hyperbolic in the standard ADM formulation.

Gauge fixing

We fix the gauge by adding a term

$$S_{\text{gauge}} = -\frac{1}{2}\int \sqrt{-g} H_a H^a$$

to the action expanded to second order in the perturbation.

For Lovelock $\rightarrow$ *harmonic gauge*

$$H_a \equiv \nabla^b h_{ab} - \frac{1}{2}\nabla_a h = 0$$

For Horndeski $\rightarrow$ *generalised harmonic gauge*

$$H_a \equiv (1+f)\left(\nabla^b h_{ab} - \frac{1}{2}\nabla_a h\right) - \mathcal{H}_a{}^b \nabla_b \psi = 0$$

Hyperbolicity of Einstein equations

Consider the linearised Einstein equations for a perturbation $g \to g + h$. In harmonic gauge

$$\nabla^b \left(h_{ab} - \frac{1}{2} g_{ab} h \right) = 0,$$

it reads

$$g^{\alpha\beta} \partial_\alpha \partial_\beta \left(h_{\mu\nu} - \frac{1}{2} g_{\mu\nu} h \right) + \ldots = 0.$$

Hyperbolicity of Einstein equations

In this gauge the principal symbol reads

$$P(\xi)^{abcd} = -\frac{1}{2}\xi^2 G^{abcd} = -\frac{1}{2}\xi^2(g^{ac}g^{bd} + g^{ad}g^{bc} - g^{ab}g^{cd})$$

where G^{abcd} is non-degenerate.

ξ characteristic iff null.

Fix ξ_i, then ξ_0 which solves $g^{\mu\nu}\xi_\mu\xi_\nu = 0$ is real $\Rightarrow$ Einstein equations are *weakly hyperbolic*.

For ξ null, all symmetric t_{ab} are eigenvectors of $P(\xi)$ (or equivalently all $(t, \xi_0 t)^T$ are eigenvectors of M).
Then M is diagonalisable $\Rightarrow$ Einstein equations are *strongly hyperbolic*.

Strong fields

In GR, equations are strongly hyperbolic in *any* background.

In Lovelock, it is known that equations fail to be even weakly hyperbolic in some backgrounds with large curvature, e.g. in exterior of small black holes (*Reall, Tanahashi, Way, 2014*).

Similarly, it is known that Horndeski theories may suffer from similar pathologies when the background fields become large (cf. gradient/ghost instabilities).

Weak fields

We will consider "weak background fields", for which the Lovelock/Horndeski equations can be considered "small perturbations" of the Einstein equations.

Example: Lovelock equations

$$G^a{}_b + \Lambda\delta^a_b + \sum_{p\geq 2} k_p \delta^{ac_1\ldots c_{2p}}_{bd_1\ldots d_{2p}} R_{c_1c_2}{}^{d_1d_2} \ldots R_{c_{2p-1}c_{2p}}{}^{d_{2p-1}d_{2p}} = 0$$

we require

$$|k_p||Riem|^p \ll 1$$

Principal symbol: $P(\xi) = P(\xi)_{\text{Einstein}} + \delta P(\xi)$

Why isn't this obviously weakly hyperbolic?

NB: the perturbation – albeit small – affects second order derivatives, hence may spoil hyperbolicity of Einstein's equations.

Example: consider the following system in 2-dim Minkowski

$$\Box\phi = k\epsilon\partial_0\partial_1\psi \qquad \Box\psi = -k\epsilon\partial_0\partial_1\phi$$

where $0 < \epsilon \ll 1$. $k \to$ "coupling", $\epsilon \to$ "field strength".

The eigenvalues of M are given by

$$\xi_0^{\pm} = \pm\xi_1\left(1 + \frac{i}{2}k\epsilon + \mathcal{O}(\epsilon^2)\right) \qquad \tilde{\xi}_0^{\pm} = \pm\xi_1\left(1 - \frac{i}{2}k\epsilon + \mathcal{O}(\epsilon^2)\right)$$

$k = 0 \Rightarrow$ hyperbolic; $k \neq 0 \Rightarrow$ elliptic.

Weak hyperbolicity

Lovelock and Horndeski theories are weakly hyperbolic in (generalised) harmonic gauge, in a generic "weak field" background.

Weak hyperbolicity - Sketch of proof [1/4]

The matrix M is hermitian w.r.t. H

$$HMH^{-1} = M^T$$

where

$$H(\xi_i) = \begin{pmatrix} B(\xi_i) & A \\ A & 0 \end{pmatrix}$$

NB. As we will see, H is *not* positive definite!

Weak hyperbolicity - Sketch of proof [2/4]

We parametrise the "deformation from GR" by ϵ.
Need to determine signature of $H(\epsilon)$.

Determine signature for GR, i.e., $H(0)$
By continuity, the signature of $H(\epsilon)$ will be the same for sufficiently small ϵ (cf. weak fields).

Weak hyperbolicity - Sketch of proof [3/4]

Residual gauge freedom yields a class of eigenvectors ("pure gauge" vectors) for $M(\epsilon)$. These arise from t_{ab} of the form

$$t_{\mu\nu} = \xi^{\pm}_{(\mu} X_{\nu)}$$

where

$$g^{\mu\nu}\xi^{\pm}_{\mu}\xi^{\pm}_{\nu} = 0.$$

The corresponding vector is null w.r.t. $H(\epsilon)$.

There are $2d$ such vectors and they form an H-null subspace. It is the *maximal* null subspace compatible with the signature of H.

Weak hyperbolicity - Sketch of proof [4/4]

Suppose $M(\epsilon)$ has a complex eigenvalue. Then the corresponding eigenvector must be null w.r.t. H! This would contradict the maximality of the null subspace spanned by the "pure gauge" vectors.

We conclude that all eigenvalues of $M(\epsilon)$ must be real and hence that *weak hyperbolicity* holds!

Strong hyperbolicity - Lovelock

The harmonic gauge linearised Lovelock EoMs are not strongly hyperbolic in a generic weak field background.

Strong hyperbolicity - Horndeski

For $\mathcal{G}_5 = \partial_X \mathcal{G}_4 = 0$ and the gauge choice

$$f = -1 + \sqrt{1 + \mathcal{G}_4}$$
$$\mathcal{H}_{ab} = \partial_\Phi f - \frac{\partial_X \mathcal{G}_3}{1 + \mathcal{G}_4} \nabla_a \Phi \nabla_b \Phi$$

the matrix M is diagonalisable and hence the linearised EoMs for the theory

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{G}_2(\Phi, X) + \mathcal{G}_3(\Phi, X)\Box\Phi + \mathcal{G}_4(\Phi)R$$

are strongly hyperbolic.

Strong hyperbolicity - Horndeski

Require gauge condition to arise as linearisation of non-linear generalised harmonic gauge condition $\Rightarrow \mathcal{G}_3 = 0$.

The most general Horndeski theory which is strongly hyperbolic in a generalised harmonic gauge, in a <u>*generic*</u> *"weak field" background is*

$$\mathcal{L}_* = (1 + \mathcal{G}_4(\Phi))R + X - V(\Phi) + \mathcal{G}_2(\Phi, X)$$

Field redefinition can set $\mathcal{G}_4 = 0$, so this is just "k-essence" coupled to Einstein gravity.

Genericity

Note that there are weak backgrounds around which Lovelock/Horndeski theories are strongly hyperbolic (e.g. exterior of large BH in Lovelock).

However these spacetimes are NOT generic.

For *generic* background, the equations are not strongly hyperbolic.

In other words, symmetry reduced eqs may be strongly hyperbolic but full eqs are not.

Dynamical violation of hyperbolicity

Strong hyperbolicity fails for generic weak backgrounds. However, for some non-generic background we can have strong hyperbolicity, e.g. FRLW solutions in Lovelock. Is this preserved by time-evolution? Linear TTT perturbations

$$-F_1(t)\ddot{h}_{ij} + F_2(t)a(t)^{-2}\Delta h_{ij} + \ldots = 0$$

$(\tilde{\xi}_0^{\pm}, \xi_i)$ characteristic iff

$$\tilde{\xi}_0^{\pm} = \pm\frac{1}{a(t)}\sqrt{\gamma^{ij}\xi_i\xi_j\frac{F_2(t)}{F_1(t)}}$$

Sign of F_2/F_1 determines hyperbolicity (cf. ghost/gradient instabilities).

Conclusions [1/3]

- Horndeski and Lovelock theories are *weakly* hyperbolic in a "weak background", in (generalised) harmonic gauge;
- Lovelock theories fail to be *strongly* hyperbolic in a generic "weak background", in harmonic gauge;
- The most general Horndeski theory which is *strongly* hyperbolic in a "weak background", in generalised harmonic gauge is "k-essence" coupled to Einstein. The gauge choice is unique.
- Any more general Horndeski theory is *not* strongly hyperbolic in any generalised harmonic gauge.

Conclusions [2/3]

Failure of strong hyperbolicity $\rightarrow$ (at best) energy estimate with loss of derivatives $\rightarrow$ non-linear problem likely ill-posed.

What else can we try?

- Modify generalised harmonic gauge condition;
- Could look at evolution equation for curvature derived from Bianchi identities: $\Box Riem = \nabla\nabla Ric + Riem^2$
- Modify ADM approach;
- Consider more restrictive class of data (e.g. Gevrey).

Conclusions [3/3]

These theories may not admit a well-posed initial value at all (or they may do just for some very restricted class of data)!

This would imply that these theories are NOT viable alternatives to GR.

Thank you!